Isoproterenol

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Continuing Education Activity

Isoproterenol is a beta-1 and beta-2 adrenergic receptor agonist indicated primarily for bradydysrhythmias. The administration and subsequent post-administration monitoring of this medication are complex and necessitate an interprofessional approach to its usage. This activity covers isoproterenol, including mechanism of action, pharmacology, adverse event profiles, eligible patient populations, contraindications, monitoring, and highlights the role of the interprofessional team in the management of isoproterenol therapy.

Objectives:

- Identify indications for the use of isoproterenol.
- Describe the mechanism of action of isoproterenol.
- Review potential complications arising from the use of isoproterenol.
- Summarize the importance of an interprofessional team approach to the use of isoproterenol.

Access free multiple choice questions on this topic.

Indications

Isoproterenol (also known as isoprenaline) is a drug used to treat bradycardia conditions. the drug has a structural resemblance to epinephrine. It first received approval for use in the United States in 1947.

Both approved and off-label indications for isoproterenol are as follows:

Isoproterenol Approved Indications

- Heart block not requiring pacing
- Cardiac arrest from heart block when pacemaker therapy is unavailable[1]

Off-label Uses

- Bradycardia
- Bronchospasm during anesthesia
- Cardiogenic shock
- Hypovolemic shock (adjunctive treatment)
- Provocation of ventricular arrhythmias in arrhythmogenic right ventricular cardiomyopathy: used during electrophysiological studies to induce ventricular arrhythmias in patients with a history of arrhythmogenic right ventricular cardiomyopathy
- Provocation of syncope during tilt table testing
- Torsades de pointes
- Beta-blocker overdose
- Ventricular arrhythmias secondary to AV block
- Short QT syndrome
- Electrical storm in patients with Brugada syndrome
- Bradycardia in a cardiac transplant patient[2][3][4][5][6][7]

Mechanism of Action

Isoproterenol is a beta-1 and beta-2 adrenergic receptor agonist resulting in the following:

- Increased heart rate
- Increased heart contractility
- Relaxation of bronchial, gastrointestinal, and uterine smooth muscle
- Peripheral vasodilation

Both beta-1 and beta-2 adrenergic receptors exert their effects through a G-alpha stimulatory second messenger system. G-protein coupled receptors are structurally composed of a seven-transmembrane-spanning protein. The extracellular domain serves as the ligand-binding site. In the inactivated state, the intracellular domain links to a G-alpha stimulatory protein bound to a GDP molecule. Upon binding of the ligand to the extracellular domain of a beta-1 receptor, the alpha subunit exchanges a GDP molecule for a GTP and becomes activated. The (now active) G-alpha protein dissociates from the intracellular domain and activates adenylate cyclase. Activated adenylate cyclase subsequently converts intracellular ATP to cAMP. The principal second messenger in this pathway, cAMP, activates protein kinase A (PKA). Activated PKA phosphorylates L-type calcium channels in cardiac myocytes, resulting in increased intracellular calcium. PKA also causes an increase in calcium release from ryanodine receptors on the sarcoplasmic reticulum.

Beta-1 adrenergic receptors are primarily concentrated in heart tissue. The terminal effects of activation of beta-1 adrenergic receptors are an increase in intracellular calcium. In cardiac pacemaker cells, increased calcium causes an increase in the slope of phase 4 of the cardiac pacemaker action potential. By increasing the slope of phase 4, pacemaker cells reach the threshold more rapidly, resulting in the characteristic increased heart rate seen in patients on an isoproterenol infusion. In non-pacemaker cardiac myocytes, an increase in intracellular calcium causes the increased contractility characteristic of isoproterenol infusion.[8]

The result of beta-1 agonism on the heart can be summarized as follows:

- Positive inotropy (contractility)
- Positive lusitropy (relaxation)
- Positive chronotropy (heart rate)
- Positive dromotropy (conduction velocity)

Beta-2 adrenergic receptors function similarly to beta-1 receptors—activation of the G-protein coupled receptor results in an increase in intracellular cAMP. The second messenger, cAMP, then activates protein kinase A (PKA). PKA phosphorylates myosin light chain kinase (MLCK), thus inactivating it. In smooth muscle cells, MLCK is responsible for the phosphorylation of myosin, leading to myosin-actin cross-bridge formation and muscle contraction. As stated, agonism of beta-2 receptors leads to inactivation of MLCK and subsequent relaxation of smooth muscle, bronchial dilation, peripheral vasodilation, and gastrointestinal and uterine smooth muscle relaxation.[9]

Other effects of isoproterenol:

- Hepatic glycogenolysis (beta-2)
- Release of glucagon from the pancreas (beta-2)
- Activation of the renin-angiotensin-aldosterone system in the kidney (beta-1)[10]

Isoproterenol is metabolized in the liver via the CYP450 enzyme system.

Administration

Isoproterenol is administered intravenously via an infusion pump. Note that adult intravenous infusion is expressed as mcg/minutes. In contrast, the pediatric infusion is expressed in mcg/kg/minute.

Available Forms

- Brand and generic: 0.2 mg/mL (1 mL, 5mL)

Adult Dosage

Bradydysrhythmias, AV Nodal Block

- 2 to 10 mcg/minute titrated to desired effect[2]

Adam-Stokes Attacks

- 2 to 20 mcg/minute IV; Start at 5 mcg/minute IV

Brugada Syndrome (off-label)

- Bolus 1 to 2 mcg followed by 0.15 to 0.3 mcg/minute for 24 hours[7]

Cardiogenic Shock (off-label)

- 2 to 20 mcg/minute continuous infusion[3]

Provocation of Syncope During Tilt Table Testing (off-label)

- 1 mcg/minute, initially, then increase based on the desired response; max dose of 5 mcg/minute

Provocation of Ventricular Arrhythmias in Arrhythmogenic Right Ventricular Cardiomyopathy (off-label)

- 45 mcg/minute for 3 minutes, then evaluate rhythm[4]

Refractory Torsades de Pointes (off-label)

- 2 to 10 mcg/minute continuous infusion titrated to patient response[6]

Pediatric Dosage

Bradycardia, AV Nodal Block

- 0.05 to 0.5 mcg/kg/minute IV, adjusted to desired effect; the max dosage is 2 mcg/kg/minute[2]

Neonatal Dosage

Bradycardia

- 0.05 to 1 mcg/kg/minute continuous infusion titrated to effect[11]

Pharmacokinetics

Isoproterenol is immediately active upon infusion. Its half-life is 2.5 to 5 minutes. Conjugation in hepatic and pulmonary tissues is the major method of metabolism. Excretion occurs via urine in the form of sulfate conjugates.

Pregnancy/Breastfeeding

The use of isoproterenol during pregnancy has not been evaluated. The presence of isoproterenol in breast milk is presently unknown.[12]

Renal and Hepatic Dosing

Renal and hepatic dosing is undefined.

Adverse Effects

Common

- Headache
- Dizziness
- Upset stomach
- Flushing
- Fatigue
- Nervousness
- Diaphoresis
- Blurred vision
- Tachyarrhythmia
- Hypertension/hypotension

Cardiovascular

- Angina
- Flushing
- Hypotension
- Hypertension
- Palpitations
- Ventricular arrhythmia
- Premature ventricular contractions
- Adams-stokes syndrome
- Bradycardia (with tilt table testing)

Respiratory

- Dyspnea
- Edema

Ophthalmic

- Blurred vision

Central Nervous System

- Headache
- Dizziness
- Nervousness
- Restlessness
- Seizures

Gastrointestinal

- Nausea
- Vomiting

Endocrine & Metabolic

- Hypokalemia
- Increased serum glucose

Musculoskeletal

- Tremor
- Weakness

Contraindications

Absolute Contraindications

- Angina
- Tachydysrhythmias
- Preexisting ventricular arrhythmias
- Digoxin intoxication
- Sulfa allergy: Contains sulfites

Isoproterenol requires caution in patients with the following:

- Cardiovascular disease: Isoproterenol causes an increase in myocardial oxygen demand
- Diabetes: May cause an increase in blood glucose levels
- Distributive shock: Beta-2 agonism will further decrease total peripheral resistance
- Hyperthyroidism: May induce thyroid storm
- The drug contains sulfites which may provoke an allergic reaction in patients with a sulfa allergy
- Elderly

Pregnancy

Isoproterenol is a Pregnancy Risk Factor C. It may interfere with uterine contractions due to its beta-2 agonist properties. Animal reproduction studies have not been conducted at this time. It is currently unknown if isoproterenol is present in breast milk; breastfeeding mothers are advised to exercise caution when taking isoproterenol. There is no human data available regarding the effect of isoproterenol on milk production.[12]

Drug Interactions

Risk C: Monitor Therapy

- Atomoxetine: Propensity to increase heart rate
- Cannabinoid-containing products: Propensity to increase heart rate
- COMT Inhibitors: Isoproterenol is degraded by catechol O-methyltransferase (COMT) and may rise to dangerous levels in the presence of a COMT inhibitor
- Doxofylline: Increased risk of doxofylline toxicity
- Tedizolid: Increased risk of a hypertensive episode

Risk D: Consider Modifying Therapy

- Topical Cocaine: Heightened risk of hypertension, tachycardia, and increased oxygen demand
- Linezolid: Increased risk of hypertension due to COMT inhibitor-like action of linezolid
- Mifepristone: QTc prolongation
- QTc prolonging agents: Avoid giving isoproterenol in combination with other QTc prolonging agents.

Risk X: Avoid or Use Alternatives

- Inhaled Anesthetics: Increased risk of arrhythmia
- Benzphetamine
- Clomipramine
- Diethylpropion
- Epinephrine
- Methamphetamine
- Midodrine
- Phendimetrazine
- Procarbazine
- Sotalol

Monitoring

Clinicians should record electrolytes, blood pressure, and ECG at baseline. Vitals (i.e., heart rate, respiratory rate, blood pressure), in addition to ECG, arterial blood gas, blood glucose levels, and serum potassium and magnesium levels, require continuous monitoring in patients receiving isoproterenol.

Toxicity

Acute toxicity from isoproterenol can cause a significant drop in blood pressure. In instances of accidental overdosage, as evidenced mainly by tachycardia or other arrhythmias, hypotension/hypertension, or angina, either reduce the rate of isoproterenol administration or discontinue the drug until the patient has stabilized. Pulse, respiration, ECG, and blood pressure should all be monitored regularly. There is no known reversal agent for humans, although studies are underway to look for agents to address toxicity.[13]

Enhancing Healthcare Team Outcomes

A cardiologist most often prescribes isoproterenol. However, the use of the drug requires an interprofessional team that consists of ICU nurses and other nursing staff, intensivists, cardiologists, cardiac surgeons, critical care specialists, and pharmacists. The drug only has an application as an intravenous drip for severe bradycardia and cardiac arrest. It is sometimes used to manage hypovolemic shock and bronchospasm. Isoproterenol can cause tachyarrhythmias and hypertension at high doses. When used in the ICU, the patient requires close monitoring. Because of the availability of pacemakers and other chronotropic drugs, the use of isoproterenol has diminished in the current medical paradigm.[14]

As stated, isoproterenol requires interprofessional collaboration for effective use. The ordering clinician (MD, DO, NP, or PA) decides to use the drug, but the entire healthcare team must be involved. This team includes specialists, as listed above, as well as pharmacists and other nursing staff.

The pharmacist needs to verify dosing and perform thorough medication reconciliation; it would be prudent to enlist the assistance of a Board Certified Cardiology pharmacist when ordering isoproterenol, as their extensive additional training can be a good resource for the team regarding interactions, dosing and dose adjustments, and advising the team on what to monitor for potential adverse events.

Nursing is at the front lines for entering patient medication history and also conducting the monitoring necessary when administering isoproterenol. Any abnormal results or concerns require communication with the team, including clinicians and the pharmacy, for dosing or drug changes. This is essential, given the nature of the conditions treated with isoproterenol and the potential adverse events and/or drug interactions, which can cause severe problems for the patient—as with pharmacy, having nursing staff with additional specialized cardiology training can prove to be of significant benefit. Only with this type of collaborative team effort and interprofessional communication can the interprofessional team optimize isoproterenol therapy, resulting in improved outcomes with fewer adverse events. [Level 5]